

Tuning Language Models by Mixture-of-Depths Ensemble

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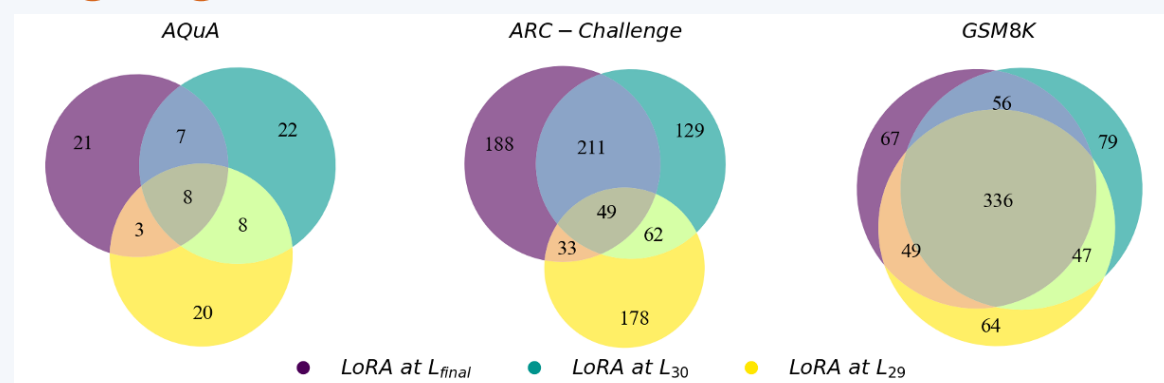
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Motivation

The **logit lens** shows late layers already carry task-relevant predictions, yet LLMs are tuned only on the *final*-layer loss. Can this interpretability observation become an **actionable training signal**?

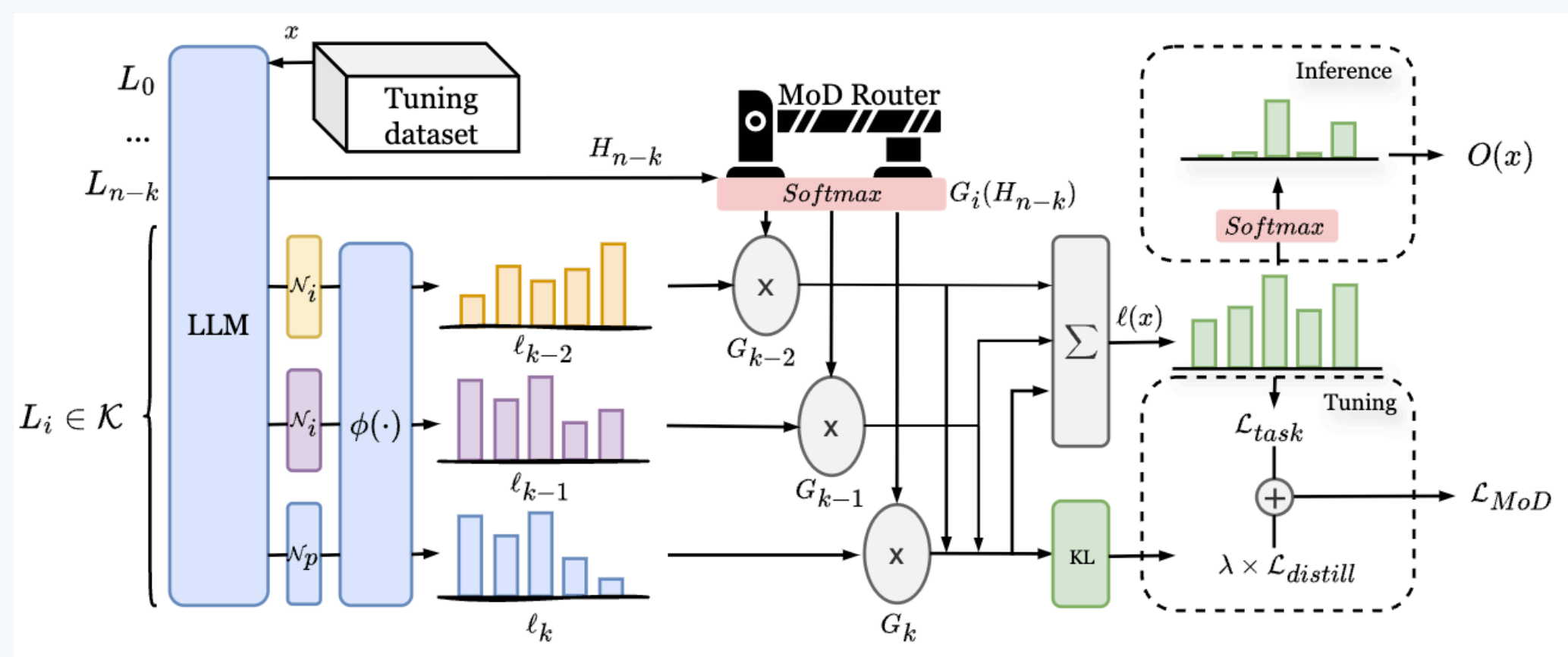


Problems solved by tuning single late layers: each solves a *complementary* set.

Take-aways

- ▶ Late-layer logits are a **trainable signal**.
- ▶ **MoDE** plugs into LoRA / DoRA at **+0.04%** params.
- ▶ Sparse routing: up to **1.6×** faster.

Mixture-of-Depths Ensemble (MoDE)



1. Early exit. LM head on each late layer $L_i \in \mathcal{K}$, with trainable norm $\mathcal{N}_i$:

$$\ell_i(x_t | x_{<t}) = \phi(\mathcal{N}_i(h_t^{(i)}))_{x_t}$$

2. Routing. A linear router mixes the k log-its:

$$\ell(x_t | x_{<t}) = \sum_{i=0}^{k-1} G(x)_i \cdot \ell_i(x)$$

3. Distillation. The final layer is the teacher:

$$\mathcal{L}_{tuned} = \mathcal{L}_{task} + \lambda \sum_{i=0}^{k-2} \text{KL}(P_i || P_n)$$

4. Sparse routing. Keep top routes for early exit:

$$G_{\text{TopK}}(x) = \text{Softmax}(\text{TopK}(x \cdot W_g))$$

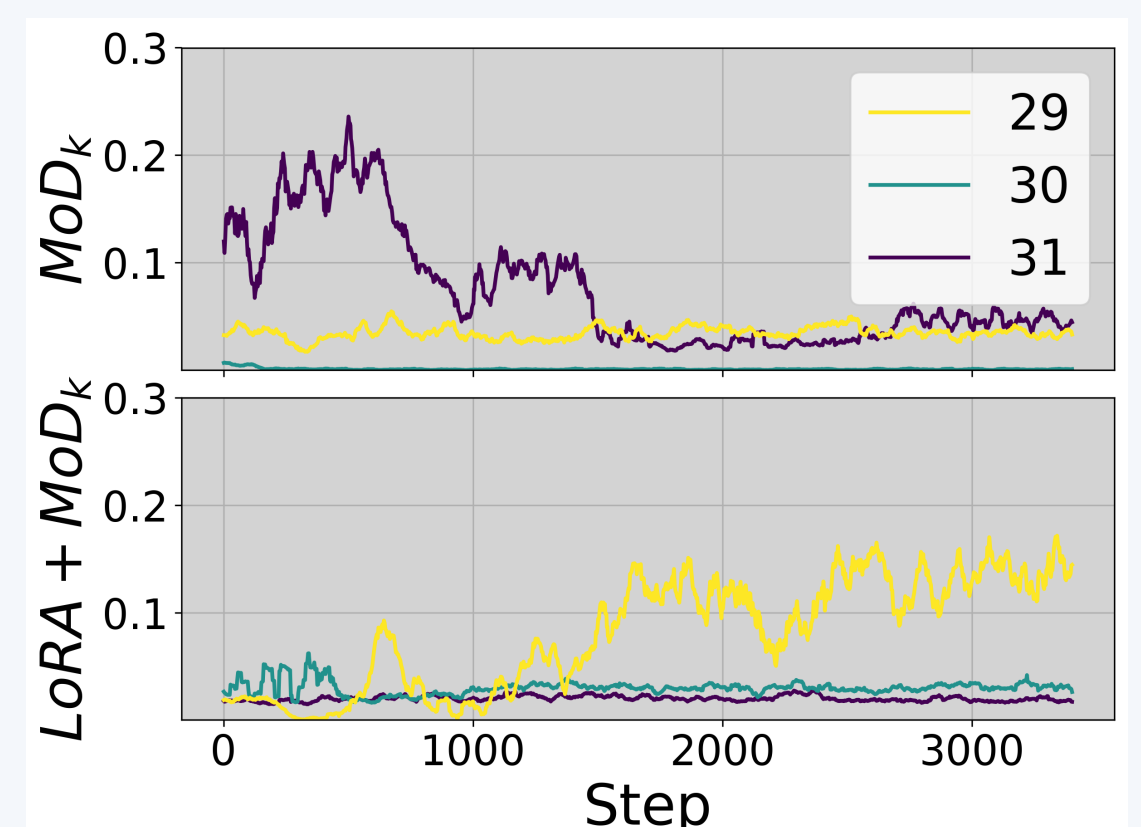
Results: Arithmetic Reasoning (LLaMA3-8B Example)

The paper evaluates LLaMA2-7B and LLaMA3-8B on arithmetic, commonsense and instruction-following tasks. As an example: LLaMA3-8B on Math7K ($k = 3$); LoRA- $\mathcal{K}$ excludes the last k layers.

Method	AddSub	AQuA	GSM8K	MAWPS	MultiArith	SingleEq	SVAMP	Avg.
LoRA- $\mathcal{K}$	91.8	25.0	60.6	90.2	98.8	95.7	72.8	76.4
LoRA _{ALL}	<u>92.7</u>	25.6	61.6	90.8	99.5	96.3	73.8	77.2
DoRA	92.2	26.8	<u>62.7</u>	91.2	98.8	<u>96.9</u>	74.0	77.5
ReFT	91.0	27.5	54.0	88.5	97.0	94.0	67.5	74.2
LoRA _{ALL} + MoDE _{$\mathcal{K}$}	93.0	26.2	62.2	<u>91.3</u>	99.5	96.5	<u>74.2</u>	<u>77.6</u>
DoRA + MoDE _{$\mathcal{K}$}	92.4	<u>27.4</u>	63.1	91.6	<u>99.0</u>	97.1	74.7	77.9
LoRA- $\mathcal{K}$ + MoDE _{$\mathcal{K}$} (+0.04%)	92.2	25.5	61.5	90.8	98.8	96.0	73.5	76.9

- ▶ MoDE **consistently improves every method it augments** (DoRA + MoDE best).
- ▶ With **minimal additional parameters (+0.04%)**, LoRA- $\mathcal{K}$ + MoDE is competitive with LoRA_{ALL} (+10.3%), showing the potential of **late-layer predictive power**.

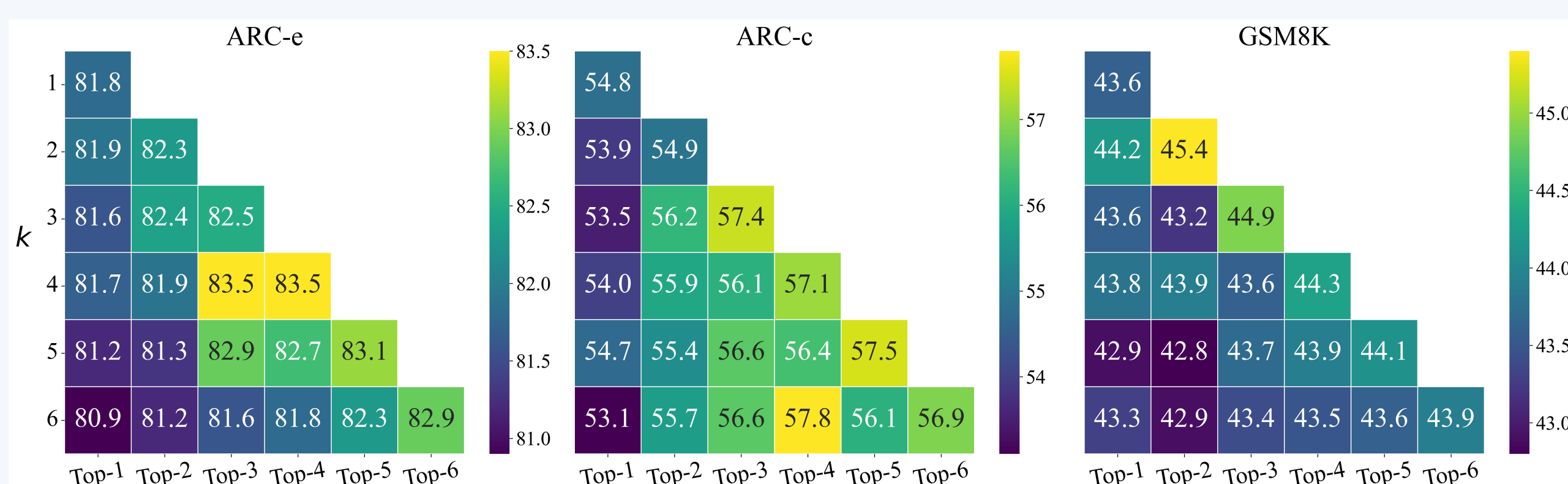
Learned Routing Pattern



Route sparsity (weights $< 10^{-5}$) on ARC-e, without (top) and with (bottom) k late-layer LoRA modules.

Without them the router relies on the last layers; with them, **earlier late layers contribute more**.

Sparse Routing: Accuracy vs. Speed



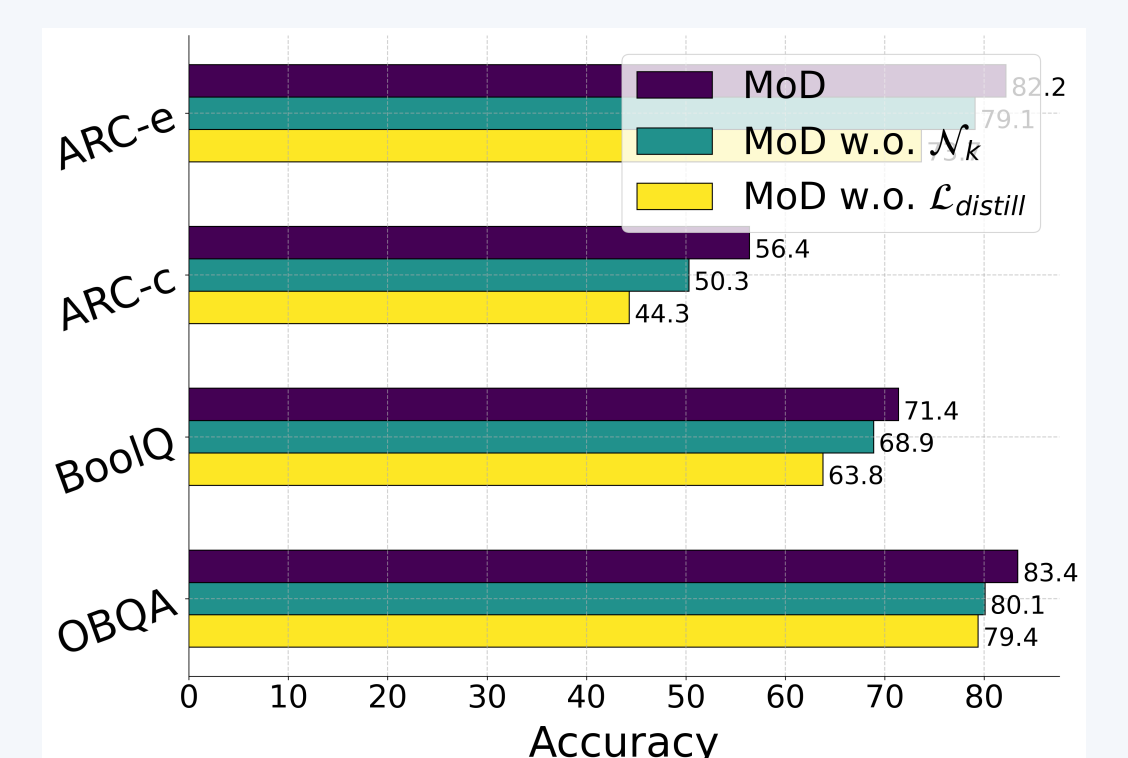
Accuracy for ensemble range k (rows) and Top- K routes (columns) on ARC-e, ARC-c and GSM8K.

- ▶ Best range is $k = 3-4$: larger k helps easy tasks (ARC-e) but hurts harder ones (GSM8K).
- ▶ Top- K costs a little accuracy but still beats $k = 1$, giving a **tunable speed trade-off**.

Speedup	Top-2	Top-3	Top-4	Top-5
ARC-e	1.4×	1.5×	1.4×	1.1×
ARC-c	1.6×	1.4×	1.3×	1.0×

Generation speedup over LoRA ($k = 6$).

Ablation



Commonsense datasets, LLaMA2-7B. w.o. $\mathcal{N}_k$: pretrained head norm; w.o. $\mathcal{L}_{distill}$: task loss only.

Both $\mathcal{L}_{distill}$ and $\mathcal{N}_k$ **enhance tuning performance**; removing either hurts. Strong last-layer task signals are key to adaptation: **distillation matters most**.